THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The midpoint of the join of $(2, 4)$ and $(6, 8)$ is: 1

(a) $(4, 7)$ **(b)** $(4, 6)$ **(c)** $(8, 12)$ **(d)** $(5, 6)$

2. In a right triangle with perpendicular 4, base 3 and hypotenuse 5, $\tan\theta =$ 1

(a) $\frac{3}{5}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{4}{5}$ **(d)** $\frac{4}{3}$

3. From a point on level ground, 5 m away from the foot of a vertical tower, the angle of elevation of its top is 60°. The height of the tower is: 1

(a) $\frac{5\sqrt{3}}{3}$ m **(b)** $10\sqrt{3}$ m **(c)** 5 m **(d)** $5\sqrt{3}$ m

4. The length of the tangent from an external point P to a circle is 8 cm and P is 10 cm from the centre. The radius of the circle is: 1

(a) 9 cm **(b)** 18 cm **(c)** 6 cm **(d)** 2 cm

5. The radius of a circle is 7 cm. The circumference of the circle is: 1

(a) 22 cm **(b)** 44 cm **(c)** 154 cm **(d)** 88 cm

6. A right circular cylinder has base radius 7 cm and height 10 cm. The volume of the cylinder is: 1

(a) 3080 cm³ **(b)** 748 cm³ **(c)** 440 cm³ **(d)** 1540 cm³

7. For the data with classes 10–20, 20–30, 30–40, 40–50 with frequencies 5, 8, 20, 17 respectively, The modal class is: 1

(a) 10–20 **(b)** 20–30 **(c)** 30–40 **(d)** 40–50

8. A die is thrown once. The probability of getting a multiple of 3 is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{6}$

9. The number of prime factors (counted with repetition) of 12 is: 1

(a) 6 **(b)** 4 **(c)** 3 **(d)** 2

10. The sum of the zeroes of $x^2 - 5x + 6$ is: 1

(a) -5 **(b)** 11 **(c)** 6 **(d)** 5

11. If $x = 2, y = 1$ is a solution of $2x + 3y = k$, then k is: 1

(a) 15 **(b)** 7 **(c)** 1 **(d)** 8

12. In $2x^2 - 3x + 5 = 0$, the value of b is: 1

(a) 5 **(b)** 2 **(c)** -3 **(d)** 3

13. The sum of the first 10 terms of $1, 2, 3, 4, \ldots$ is: 1

(a) 66 **(b)** 110 **(c)** 45 **(d)** 55

14. In $\triangle ABC$, $DE \parallel BC$, $AD = 2$, $DB = 3$, $AE = 4$. Then $EC =$ 1

(a) 4 **(b)** 3 **(c)** 6 **(d)** 8

15. The point $(3, 4)$ lies in the: 1

(a) third quadrant **(b)** fourth quadrant **(c)** first quadrant **(d)** second quadrant

16. The value of $\sin 30^\circ$ is: 1

(a) 0 **(b)** 2 **(c)** 1 **(d)** $\frac{1}{2}$

17. From a point on level ground, 9 m away from the foot of a vertical tower, the angle of elevation of its top is 30°. The height of the tower is: 1

(a) 9 m **(b)** 3 m **(c)** $3\sqrt{3}$ m **(d)** $9\sqrt{3}$ m

18. A tangent to a circle of radius 7 cm has length 24 cm from the external point P to the point of contact. The distance of P from the centre of the circle is: 1

(a) 25 cm **(b)** 50 cm **(c)** 31 cm **(d)** 17 cm

19. The circumference of a circle is 44 cm. The radius of the circle is: 1

(a) 7 cm **(b)** 3.5 cm **(c)** 14 cm **(d)** 22 cm

20. A right circular cone has base radius 7 cm and height 24 cm. The slant height of the cone is: 1

(a) 25 cm **(b)** 15.5 cm **(c)** 23 cm **(d)** 31 cm

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Find the distance between the points $(0,\ 0)$ and $(3,\ 4)$. 2

22. In a right triangle the perpendicular is 4 and the hypotenuse is 5. Find $\sin\theta$. 2

23. From a point on level ground, 10 m away from the foot of a vertical tower, the angle of elevation of its top is 45°. Find the height of the tower. 2

24. A point P is 25 cm from the centre O of a circle of radius 20 cm. Find the length of the tangent drawn from P to the circle. 2

25. The radius of a circle is 3.5 m. Find the area of the circle. 2

26. A right circular cylinder has base radius 7 cm and height 10 cm. Find the total surface area of the cylinder. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. A ladder 12 m long leans against a vertical wall and makes an angle of 60° with the level ground. Find the height reached on the wall and the distance of the foot of the ladder from the wall (in this order). 3

28. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 12 cm. The length of each tangent is 16 cm. Find the perimeter of the quadrilateral $PAOB$. 3

29. A sector of a circle of radius 21 cm has central angle 45°. Find the length of its arc. 3

30. A right circular cone has base radius 7 cm and height 24 cm. Find the volume of the cone. 3

31. The marks obtained by students in a test are given below. 3

Marks	0–10	10–20	20–30	30–40	40–50
Number of students	12	20	8	4	6

Find the mean.

OR

In a lottery there are 10 prizes and 25 blanks. A ticket is drawn at random. Find the probability of getting a prize. 3

32. Verify that HCF $\times$ LCM $=$ product of the numbers for 12 and 30. 3

OR

Find a quadratic polynomial whose sum and product of zeroes are 5 and 6 respectively. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. Tangents PA and PB are drawn from a point P at a distance of 5 cm from the centre O of a circle of radius 3 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). 5

34. A circular park of radius 14 m is surrounded by a path 7 m wide. Find the area of the path. 5

35. A solid metal cuboid 24 cm by 12 cm by 6 cm is melted and recast into cubes of edge 3 cm. Find the number of cubes obtained. 5

36. The marks obtained by students in a test are given below. 5

Marks	50–60	60–70	70–80	80–90	90–100	100–110
Number of students	14	2	20	2	9	3

Find the mean.

OR

A bag contains 3 red, 4 blue, 3 green balls. One ball is drawn at random. Find the probability that the ball is:
(i) red,
(ii) not blue,
(iii) yellow (in this order). 5

37. Find the LCM and HCF of 12, 18 and 24 using prime factorisation. 5

OR

Verify the relationship between zeroes and coefficients for the cubic $x^3 - 6x^2 + 11x - 6$ whose zeroes are $1, 2, 3$. 5

38. 2 pencils and 3 erasers cost ₹22, while 3 pencils and 2 erasers cost ₹23. Find the cost of one pencil and one eraser. 5

OR

The length of a rectangular plot exceeds its breadth by 5 m and its area is 66 m². Find its dimensions. 5

END OF QUESTION PAPER Total: 80 Marks